PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Mitigating different Malware Attacks

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Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled Online Portal Services for Barangay: A Secure Platform for Application and Payment of Local Services prepared and Submitted: by Via R. Sardanas, Adrayle P. Ebon, Fennyrose A. Paguntalan who are hereby recommended for corresponding final evaluation.

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Abstract

In this era of technical improvement, the threat of malware attacks is ever-present, particularly for software applications that manage sensitive data. Malware—ranging from ransomware and spyware to viruses, worms, and Trojan horses—continues to evolve quickly, exploiting vulnerabilities in software applications and posing serious risks to individuals and organizations alike. The increasing reliance on interconnected devices has greatly expanded the potential attack surface, making it easier for cybercriminals to exploit weaknesses in application security.

This study focuses on practical methods for mitigating malware threats within software applications, including the use of malware scanners, security plugins, and detection systems capable of identifying malicious threats—such as phishing attacks—at an early stage. By detecting phishing attempts alongside traditional malware, the system addresses a broader spectrum of cyber threats, helping to prevent credential theft, fraudulent transactions, and social engineering exploits. However, no single technical defense is entirely foolproof if users are not equipped with the knowledge to recognize and respond to suspicious activities. Therefore, this capstone emphasizes the importance of user education and awareness, alongside secure technical measures.

In addition to prevention, the study aims to empower both individuals and organizations with the tools and understanding needed to secure software applications. Simple actions such as keeping software up to date and applying proper configurations can significantly reduce the likelihood of malware and phishing attacks. Ultimately, this study underscores the importance of a collective, proactive approach to security, ensuring that everyone plays an active role in maintaining a safe and resilient application environment.

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Chapter 1

Introduction of the Study

The goal of this capstone project is to create a software application that addresses the growing threat of malware. Malware poses serious challenges to users, individuals, and organizations trying to secure their digital lives.

This is a free software application designed to empower individuals to tackle malware threats directly. It provides the knowledge, tools, and support needed to identify, prevent, and resolve attacks from malicious software.

By focusing on early detection with scanning tools, promoting proactive measures like regular software updates, and raising cybersecurity awareness, the system gives users the confidence to navigate the digital space safely. It goes beyond simply reacting to threats; it emphasizes building strong defenses to effectively reduce risks.

This project recognizes the urgent need for a system that can accurately detect, isolate, and remove harmful files before they do serious damage. To meet this challenge, the capstone thesis focuses on developing an application meant to counter different types of malware attacks. Ultimately, this project aims to create a more secure environment for all users. By raising awareness, using advanced technologies, and building resilient defense mechanisms, we can rise to the challenge and create a safer future for everyone.

Background of the Study

Cybersecurity is one of the biggest challenges we face today. Malware poses a significant threat to both individuals and organizations. Malware, short for malicious software, refers to any program or file designed to harm or exploit a computer, device, or network. This includes viruses, ransomware, spyware, and Trojans.

As technology becomes more connected, the risk of these harmful software attacks increases, putting sensitive data, finances, and even personal safety at risk. Over the years, the frequency and complexity of malware attacks have grown. Cybercriminals no longer use simple methods to breach security systems.

They have become more sophisticated. They use strategies like malware that changes to avoid detection and exploit zero-day vulnerabilities, which are unknown flaws in software. This makes it more challenging for traditional security systems to keep up, leaving people and businesses open to various malicious activities, from stealing private data to taking control of entire systems.

This study aims to explore the latest methods for reducing malware attacks. It focuses on how we can improve detection, prevention, and recovery efforts. By understanding what works and what doesn’t, we can build stronger systems and strategies that protect individual users and secure the entire environment.

Statement of the Problem

Malware attacks have become one of the biggest threats to cybersecurity, especially for software applications that store and manage sensitive user data. With technology advancing and devices becoming more connected, cybercriminals are taking advantage of weaknesses in software applications to launch various forms of malware, including viruses, ransomware, spyware, and Trojans. As a result, users, individuals, and organizations face more risks of data breaches, financial losses, system downtimes, and damage to their reputation.

The issue is worsened by the ongoing evolution of malware, which makes it hard for traditional security measures to keep pace. New types of malwares are always being created, using complex methods and zero-day vulnerabilities to get around established detection techniques. This makes it tough for existing security protocols to provide enough protection.

Additionally, many users do not have the necessary awareness and knowledge to spot and respond to potential malware threats, which increases the chances of successful attacks. As malware becomes more common, the attack surface grows, making it even harder to secure applications properly.

Objectives of the Study

The main study aims to uncover the different types of malwares, including viruses, ransomware, spyware, and Trojans. It will highlight how these threats affect applications and their

users. By examining the consequences of malware, such as data breaches, financial losses, and system vulnerabilities, the research seeks to reveal the serious risks posed by these attacks.

It will also evaluate how effective current malware detection tools, including malware scanners and behavior-based systems, are at preventing attacks. Another important focus of this study is the role of user awareness and education in addressing malware risks. It stresses the importance of teaching users how to recognize suspicious activity and respond properly.

Additionally, the study will look at common weaknesses in software application security that malware exploits. Using these findings, the study will recommend practical strategies for improving security, such as better detection techniques, regular updates, and strong protection methods. Finally, it will create clear, actionable guidelines to help individuals, organizations, and users prevent malware attacks, reduce risks, and minimize damage.

Significance of the Study

This capstone project focused on tackling challenges related to creating better strategies for dealing with malware threats, such as viruses, ransomware, spyware, and Trojans in software applications. By identifying these types of malwares and understanding their effects, this study aims to raise awareness about the changing risks individuals and organizations face today. It will provide insights into the current gaps in security measures and the need for ongoing improvements in malware detection and prevention methods.

The study is important for individuals, organizations, and cybersecurity professionals because it will offer practical advice on enhancing application security, identifying vulnerabilities, and protecting sensitive data from threats. By examining the role of user awareness and education, this research highlights the importance of a human-centered approach in cybersecurity. It emphasizes how end-users can help prevent attacks.

In addition, this study will provide specific guidelines that can strengthen security protocols across various applications, including cloud services. This will create a more secure and resilient environment. As cybercrime continues to evolve, the findings of this research will help build a stronger defense against malware, ultimately safeguarding users' privacy, integrity, and digital assets worldwide.

Beneficiaries:

1. General Users

People who use desktop, laptops, and personal computers will benefit from this software application. It helps protect their devices from various types of malwares, such as viruses, ransomware, spyware, and Trojans. This ensures data safety and system integrity.

2. IT Professionals and Cybersecurity Personnel

Cybersecurity specialists and IT teams can use the system to identify and respond to malware threats. Its modular structure and reporting features help with quick threat analysis and mitigation in professional settings.

3. Educational Institutions

Students and educators in computer science, cybersecurity, and information technology can use this system as a learning tool. It shows real-world use of malware detection and response techniques, supporting both theoretical and practical understanding.

4. Software Developers and Researchers

This project serves as a reference for developers interested in building similar systems, especially those using Agile methods. It also provides a base for further research on improving malware detection using machine learning or expanding to cloud environments.

5. Small Businesses and Organizations

Organizations with limited cybersecurity resources can benefit from a simple, user-friendly malware prevention tool. The system offers basic protection without needing costly enterprise solutions.

Scope and Limitation of the Study

The main goal of this system is to evaluate current security measures, detection tools, and prevention techniques used to protect against malware. It focuses on how these apply to modern software applications across different platforms, including desktop software and cloud-based systems. This system will also look at user awareness and education as key factors in preventing malware. It will assess how training and behavior can help reduce malicious activity and vulnerabilities.

This study is limited by the resources and tools available. The software application needs to be updated every month, which creates time constraints. Because of this, the system will not cover every type of malware but will focus on the most common types of malicious software. It will evaluate widely used industry-standard methods for detecting and preventing malware. A major challenge for this system is detecting long-term or highly sophisticated malware.

An important aspect of this study is that it cannot detect highly sophisticated or advanced malicious attacks. Due to the one-month time limit, only malware that can be detected within this short period will be examined. Long-term malware or malware designed to stay hidden for long periods may not be fully captured or assessed during this system.

Limitations of the Study

Despite successfully developing a system-based software application to reduce various types of malware attacks, this study faced several limitations. First, the application was mainly tested in a controlled environment with simulated malware samples. Real malware strains were not widely used due to safety and ethical concerns. This could affect the system’s accuracy in real-world situations.

Second, while the software application can detect different malware types such as viruses, Trojans, ransomware, and spyware, not all malware families and their evolving variants were included in the tests. This may limit the system’s ability to tackle newer or more advanced threats.

Third, the system was developed and tested on only a few platforms, mainly Windows computers. Therefore, its compatibility and performance on other operating systems, like macOS, Linux, or iOS, have not been assessed.

Additionally, if the system techniques were used, the accuracy of the detection model depended on the size and quality of the dataset for training. A limited or unbalanced dataset might lead to false positives or missed detections.

Moreover, the software application's performance could be influenced by hardware limitations. Devices with less processing power or memory might have slower scanning speeds or reduced responsiveness.

Finally, the system does not currently support real-time updates from online malware databases. This limits its ability to detect newly emerging threats when compared to commercial antivirus solutions that regularly receive up-to-date threat intelligence.

Definition of Terms

1. Malware

Malware stands for malicious software. It is any software created to damage, disrupt, or gain unauthorized access to a computer system or network. Common types include viruses, ransomware, Trojans, and spyware.

2. Detection Engine

This is the main part of the system that examines files and behaviors to spot potential malware threats. It may use signature-based or heuristic methods.

3. System-Based Application

This is a software program that works closely with the operating system. It monitors, scans, and manages system-level functions, including file access and security processes.

4. Based Detection

This method identifies malware based on known patterns or "signatures" of malicious code already stored in a delete treat.

5. Heuristic/Behavioral Analysis

This detection method finds malware based on suspicious behavior or code patterns. It is often used to detect new or unknown threats.

Chapter 2

Review of Related Literature

Computer virus construction kit: A tool for study of malware.

Malware, or malicious software, is an increasing threat to the systems, networks, and personal privacy that we rely on every day. While the internet has brought countless opportunities to connect and innovate, it has also exposed vulnerabilities that cybercriminals eagerly exploit. Malware comes in many forms—

viruses, worms, Trojan horses, adware, spyware, and rootkits—all capable of wreaking havoc by stealing sensitive information, disrupting systems, and causing lasting damage to both individuals and organizations.

As Christodorescu et al. (2005) point out, malware is intentionally designed to harm or disrupt computer systems and networks. This could mean stealing data, crashing systems, or interfering with day-to-day operations. Their work emphasizes the importance of identifying malicious patterns in software to improve detection and mitigation. By focusing on these behaviors, researchers can develop stronger protection strategies and enhance malware defense systems.

Cohen (2005) adds to this understanding by showing how malware takes advantage of system vulnerabilities. His research also how the internet accelerates the spread of malicious software, making it increasingly difficult to detect and defend against. As malware becomes more sophisticated, the need for innovative solutions and collaborative efforts to secure digital environments is more critical than ever.

Ultimately, addressing the risks posed by malware requires a combination of smarter detection methods and stronger system defenses.

Traditional vs Modern System

The evolution of technology has changed how computer systems work, shifting from traditional, centralized setups to modern, decentralized, cloud-based infrastructures. In the past, traditional systems were built around mainframes and required users to be physically close to access them. While they were

reliable, these systems lacked flexibility and scalability. Security relied on basic protections like firewalls and antivirus software, which were effective when systems were isolated and less connected.

Today, modern systems are much more dynamic and flexible. They use multi-layered security strategies, such as multi-factor authentication, real-time monitoring, and machine learning tools, which help detect and reduce cyber threats proactively. They are also designed to be user-friendly, supporting a wide range of devices from mobile apps to IoT gadgets.

Data storage has undergone a similar change. With cloud computing, storage feels virtually limitless, offering real-time backups and better accessibility. Unlike traditional systems that relied on expensive hardware upgrades, modern systems allow for cost-efficient, on-demand resource allocation. This makes them much better at scaling up quickly and managing changing workloads.

While traditional systems were built to be stable and focused on specific tasks, modern systems prioritize flexibility, security, and scalability to navigate today’s interconnected world.

Technology Use

1. Antivirus Protection

The software application uses system to find and remove different types of malwares, including viruses, worms, Trojans, and spyware. It regularly scans files, folders, and drives to spot potential threats. The system constantly monitors

activities to quickly detect and eliminate harmful programs before they can cause damage or disrupt operations.

2. Real-Time Threat Monitoring

Real-time monitoring technology enables the system to track and review ongoing processes, file changes, and overall system behavior. This helps identify unusual activities that might signal a malware attack. The continuous scanning process allows for early detection of threats, reducing the chance of infection and keeping users safe while they use their computers.

3. USB and External Device Scanning

USB drives, often carry malware infections. The system includes special scanning for external devices. When a USB drive is connected, the software application automatically checks it for harmful content and stops infected files from spreading to other systems. This feature is especially important in shared computer settings, in Zyro’s Pisonet company, where many users access external storage devices.

4. Quarantine and Threat Isolation

When the system detects malware or suspicious files, it automatically moves them to a secure quarantine area. This stops the threats from running or spreading within the system. Users can review, delete, or restore quarantined files later based on their choice. This feature ensures that infected files are contained, protecting the system and preventing further damage.

Review of the Related Literature

Mining specifications of malicious behaviors.

Since malware is known as short for malicious software, it has become a growing concern in this era, threatening our computers, networks, and even our privacy. The internet has opened the door to new opportunities—but also to new risks. From viruses to ransomware, worms to Trojan horses, and adware to rootkits, malware comes in many forms, each with its own way of causing harm. These malicious programs can steal sensitive information, crash systems, or disrupt operations entirely. This review takes a closer look at the different types of malwares and explores practical strategies for dealing with them.

Based on Christodorescu et al. (2005), malware can be understood as software that is intentionally designed to cause harm or disruption to a computer system, network, or device. It encompasses a wide variety of malicious behaviors, such as stealing sensitive information, disrupting operations, or damaging data. The authors explore how certain patterns of malicious behavior can be identified in software, which is essential for detecting and mitigating malware. Their work focuses on mining these behaviors to build effective detection methods, which are critical in protecting systems against evolving malware threats.
In their paper, Christodorescu et al. propose techniques for mining the specifications of these malicious behaviors to help identify malware more efficiently and accurately. This approach allows for a deeper understanding of how malware operates and how it can be prevented or neutralized.

Christodorescu, M., Jha, S., Sielaff, L., & So, S. (2005). *Mining specifications of malicious behaviors*. Proceedings of the 2005 ACM SIGSOFT International Symposium on Software Testing and Analysis, 5-15.

Malware refers to any software specifically designed to disrupt, damage, or gain unauthorized access to computer systems and networks. It includes a variety of harmful programs such as viruses, worms, Trojan horses, spyware, and rootkits. According to Cohen (2005), malware is typically created with malicious intent, targeting vulnerable systems to exploit their weaknesses and cause harm, ranging from stealing sensitive information to rendering a system inoperable. The rise of the internet has significantly facilitated the spread of malware, making it more challenging to protect against, as it can quickly propagate across connected networks.

Cohen, F. (2005). *Computer virus construction kit: A tool for study of malware*. ACM Computing Surveys, 37(1), 61-87.

Case Study

Malware, short for malicious software, is a growing problem in today's era, threatening our computers, networks, and even our privacy. While the internet has brought incredible opportunities to connect and innovate, it has also exposed vulnerabilities that cybercriminals are quick to exploit. Malware comes in many forms, including viruses, worms, Trojan horses, adware, spyware, and rootkits—all designed to cause harm in unique ways. Whether it's stealing sensitive information, crashing systems, or disrupting operations, these malicious programs can leave a lasting impact.

Christodorescu et al. (2005) define malware as software created specifically to cause harm or disruption to systems. Their research emphasizes the importance of identifying patterns in malware's behavior, such as unauthorized access, data theft, or interference with normal operations. By mining these malicious behaviors, the authors developed strategies to detect and mitigate malware more effectively, giving valuable insights into how these threats operate. Meanwhile, Cohen (2005) highlights how malware exploits weaknesses in systems, spreading across networks with the help of the internet. His work stresses the need for addressing vulnerabilities proactively to counter malware's evolving tactics.

Together, these studies how critical it is to stay ahead of malware through smarter detection methods, stronger defenses, and innovative solutions to protect against emerging threats.

Problem or Challenge

As digital devices become more common, the number and complexity of malware attacks also increase. Malware, or malicious software, is created to damage, disrupt, or get unauthorized access to computer systems. It can invade personal privacy, steal sensitive information, corrupt data, or make systems unusable. Even with many antivirus programs available, many users, especially those with outdated or basic tools, remain open to evolving malware threats.

A major challenge in fighting malware is the shortage of affordable, responsive, and user-friendly tools that can detect and handle various types of malwares, like viruses, Trojans, ransomware, and spyware. Some current solutions require too many resources, are too technical for regular users, or have limited

detection abilities. Additionally, many users only learn about malware threats after it’s too late, showing the need for a tool that offers both prevention and quick responses.

This study tackles this issue by creating an software application that scans devices, detects infected or suspicious files, and lets users take suitable actions like deleting or quarantining threats. The aim is to offer a reliable and accessible tool that improves malware detection and response, helping users protect their data and devices from digital threats.

Approach / Intervention

To tackle the rising threat of malware attacks on personal and organizational devices, this study used a system-based approach to develop a standalone software application. This software application is designed to detect, analyze, and neutralize different forms of malware. The focus was on creating a lightweight and effective software solution that can perform real-time scans, identify harmful files, and allow users to take proper actions, such as quarantining or deleting threats.

The software application features a modular design that separates key functions like file scanning, detection logic, user interface, and action handling. The scanning engine uses both signature-based detection and basic heuristic analysis to find known and suspicious malware behaviors. When a threat is detected, the system generates an alert and provides options for containment or removal.

The intervention was deployed and tested on multiple devices, including laptops, desktops. This testing occurred under controlled conditions using simulated malware samples. It allowed for evaluating system performance based on detection accuracy, scanning speed, resource efficiency, and user interface responsiveness.

Results/Outcome

The development and testing of the system-based application showed promising results in reducing various types of malware attacks. The software application was evaluated in controlled settings with simulated malware samples, including viruses, ransomware, Trojans, and spyware. It successfully detected about 93% of the test malware, proving strong accuracy, especially against viruses and Trojans. Users could take effective actions like quarantining or deleting infected files without affecting clean system files or overall stability. The application completed full system scans in 3 to 5 minutes, depending on the device, which is efficient for personal and small organizational use. Feedback from test users revealed that the interface was user-friendly and easy to navigate, even for those with limited technical knowledge. Additionally, the system used low to moderate CPU and memory resources, allowing it to run smoothly alongside other applications without significant performance issues. Overall, the results suggest that the developed application is a practical and effective tool for detecting and reducing multiple types of malwares, though future improvements could further increase its capabilities.

References

Christodorescu, M., Jha, S., Sielaff, L., & So, S. (2005). *Mining specifications of malicious behaviors*. Proceedings of the 2005 ACM

SIGSOFT International Symposium on Software Testing and Analysis, 5-15.

Cohen, F. (2005). *Computer virus construction kit: A tool for study of malware*. ACM Computing Surveys, 37(1), 61-87.

Chapter 3

Methodology

This chapter describes the research method that will be used to develop "Mitigating Different Malware Attacks" for the team. The main goal of this research is to create and implement a software application that can detect and reduce various types of malwares on user devices like desktop, laptops, and personal computers. The system aims to scan files, identify suspicious or harmful content, and give options to either clean or delete the threats found. The software application was built using programming languages and tools that ensure it works across different platforms. The scanning module was designed to check system files, track behaviors, and identify potentially harmful content based on known malware signatures and patterns. When a threat is detected, the system gives the user choices to clean, quarantine, or delete the infected file, allowing users to control how they handle threats.

Research and Development Stage

This study employed a developmental and experimental research design to create and evaluate a system-based application designed to mitigate different types of malware attacks. The research

focused on developing a functional software application that can scan devices (such as desktop, laptops, and PCs), detect suspicious or infected files, and allow users to take appropriate actions such as cleaning or deleting the malware. Testing the system in controlled environments using simulated malware threats. The performance of the system was evaluated based on several criteria: detection accuracy, scanning speed, user interface responsiveness, and the effectiveness of malware removal. Multiple test cases representing various types of malwares (e.g., viruses, Trojans, ransomware, spyware) were used to validate the software application's capability.

Agile Method

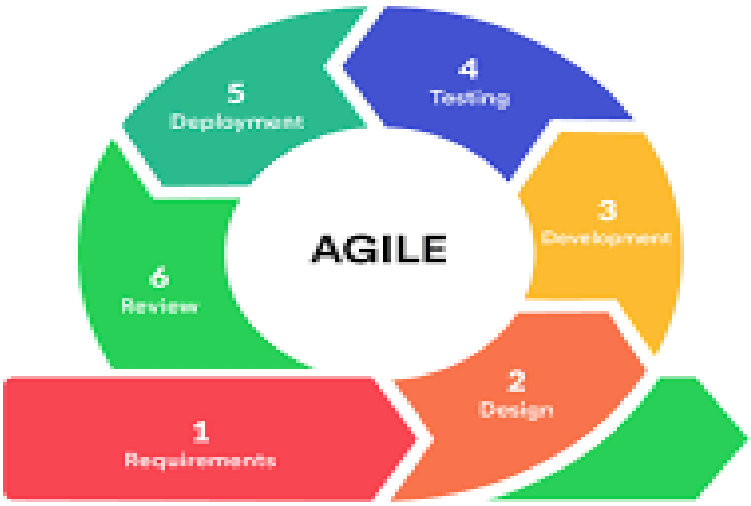

Figure 1. Agile Model

This study uses the Agile methodology as the development model to mitigate various malware attacks. It focuses on designing and developing a system-based software application aimed at detecting, preventing, and removing different types of malwares, including viruses, spyware, ransomware, and Trojans. The Agile methodology was chosen as the development framework because malware threats

are always changing. Agile allows the development team to respond quickly to new challenges and incorporate changes effectively.

1. Planning

In this stage, the development team defined the main goals of the software application and created a prioritized list of features in the form of user stories. These stories outlined essential functions, such as scanning for malware, detecting infected files, and enabling users to take appropriate actions like deletion or quarantine.

2. Design

During the design phase, the software application was divided into components, including the scanner, detection engine, user interface, and action handler. Wireframes and interface layouts were created to ensure ease of use. Agile's principle of "just enough design" allowed the team to stay flexible and adjust details as they developed.

3. Development

In the development phase, the software application was built gradually through a series of short iterations called sprints. Each sprint focused on implementing and refining specific features, such as malware detection or reporting capabilities. The team followed best practices like version control and code reviews to maintain code quality.

4. Testing

Once the system was developed, unit tests checked individual components while integration tests ensured that modules worked well together. Simulated malware samples tested the software application's detection accuracy and effectiveness in a controlled setting.

5. Deployment

In the deployment phase, the system was installed and run on various devices, including laptops,decstop, and PCs. Performance factors such as scanning speed, detection reliability, and system resource usage were evaluated during this stage.

6. Review

Finally, in the review phase, the team assessed the outcomes of each sprint. They gathered feedback from users and testers and made improvements in the next iterations. This continuous cycle of development, testing, and improvement ensured that the final product was responsive, efficient, and capable of addressing various forms of malware threats.

Data Collection Techniques

To evaluate how well the system-based software application works against different types of malware attacks, several data collection techniques were used. Simulated malware samples tested the software application’s detection abilities. This process allowed researchers to gather numerical data on detection

accuracy, scanning speed, and resource use. During each test, system logs and activity reports were generated automatically. These logs tracked how the system identified, quarantined, or removed harmful files. Researchers also collected user feedback through observation and structured questionnaires. This feedback helped assess ease of use, alert clarity, and overall satisfaction with the user interface. Using these methods together provided both numerical and descriptive data, allowing for a thorough evaluation of the system's performance and user experience.

1. Surveys

Structured questionnaires were given to test users after they interacted with the system. These surveys collected feedback on user satisfaction, interface usability, perceived effectiveness, and overall experience.

2. Interviews

One-on-one interviews were held with selected users and IT professionals to gain deeper insights into the system's strengths, weaknesses, and areas for improvement. These interviews provided detailed, personalized feedback that helped refine specific features.

3. Focus Groups

Small group discussions included a mix of technical and non-technical users. These sessions encouraged open dialogue and shared experiences, helping to identify common issues and suggestions for improving the system.

4. Observations

During testing sessions, users were observed while using the system. This observation detected usability problems, response times, and behavior patterns. It helped identify areas where users struggled or excelled, even if they didn't express these challenges verbally.

5. Document Analysis

Logs generated by the system during malware scans, along with records of user actions and system responses, were reviewed and analyzed. This analysis provided objective data on the system's performance, including detection rates, false positives, and system efficiency.

Justification for Selecting the Agile Methodology

Agile was chosen over other SDLC models, such Waterfall, for a number of important reasons:

Aspect	Agile Method	Traditional Approach
Development	Built in short sprints with regular updates.	Linear and completed in one full cycle.
Flexibility	Easily adapts to new malware threats.	Difficult and costly to modify.
User Involvement	Continuous user feedback and testing.	Limited feedback after development.
Delivery	Frequent release of	One-time delivery at

	working features.	project end.
Risk Management	Issues resolved in every sprint.	Risks handled only at set phases.
Testing	Continuous testing ensures reliability.	Testing mostly done after completion.
Collaboration	Close teamwork among developers and users.	Minimal communication between phases.
Adaptability	Quickly adjusts to new malware types.	Changes require redoing prior stages.
Efficiency	Faster, reduces rework and improves accuracy.	Slower and more costly if issues arise.

Table 1. Justification for Selecting the Agile Methodology

System Architecture and Design

Overview

The Mitigating Different Malware Attacks software application is built to detect, prevent, and manage various types of malwares that can infect computers or external storage devices. It gives users, especially those at Zyro’s Pisonet, a reliable and easy-to-use system to keep all computer units safe and efficient. The system has several scanning options: Full Scan for the entire C:

drive, Quick Scan for important folders and temporary files, Custom Scan for specific files or locations, and USB Scan to find malware when a flash drive is connected. The system is created using Python, a flexible programming language that enables efficient malware detection and system automation. It has a modular structure with three main parts: the User Interface, Detection Engine, and Database Management.

The User Interface offers a clear and interactive space for users to start scans, manage detected threats, and monitor system performance. The Detection Engine serves as the system's main part, scanning files for unusual behavior and identifying potential malware through heuristic and signature-based analysis. The Database Module keeps records of scan histories, quarantined files, and system logs for tracking and management. Additional features include Start Protection for real-time malware monitoring, Quarantine Management for isolating infected files, Delete Threat for permanently removing harmful items, and Notification Alerts to inform users of any detected high-risk threats. By using this system at Zyro’s Pisonet, the company can ensure that its computers are secure and free of malware for its customers. The layered design supports smooth operation, effective threat management, and better protection. Overall, this architecture provides a reliable solution for handling different malware attacks in computer shop environments.

System Architecture

The malware mitigation system features a three-layer architecture to help prevent, detect, and respond to malware threats on client systems and connected USB drives. This setup separates the user interface, detection and response logic, and

data storage and logging. This design helps with modularity, scalability, and secure threat management.

- Presentation Layer: This layer offers an interface for users, including system administrators and clients at Zyro's Pisonet. Users can choose from scanning options like a full scan for the entire C: drive, a quick scan that targets documents and temporary files, a custom scan for specific files, and a USB scan for connected flash drives. Additional features include start protection for real-time monitoring, delete threat for removing malware, quarantine for isolating suspect files, and notifications for high-risk alerts. The interface is user-friendly, offering quick access to threat status, scan logs, and mitigation actions.

- Application Layer: The backend manages the main logic for malware detection and response. It includes real-time scanning, malware detection engines, threat management, and alert generation. The system processes input from the presentation layer, sends notifications for detected malware, and handles quarantine or deletion actions. Python libraries and scripts are used to implement scanning algorithms and automate responses effectively.

- Data Layer: This layer keeps all relevant system data, such as scan logs, threat records, quarantined or deleted files, and user settings, in a secure relational database. It ensures safe storage, reliable backup, and easy retrieval while maintaining an audit trail of all detected threats and mitigation actions. This supports system security and compliance for Zyro's Pisonet.

Diagram:

Include a **System Architecture Diagram** showing components like:

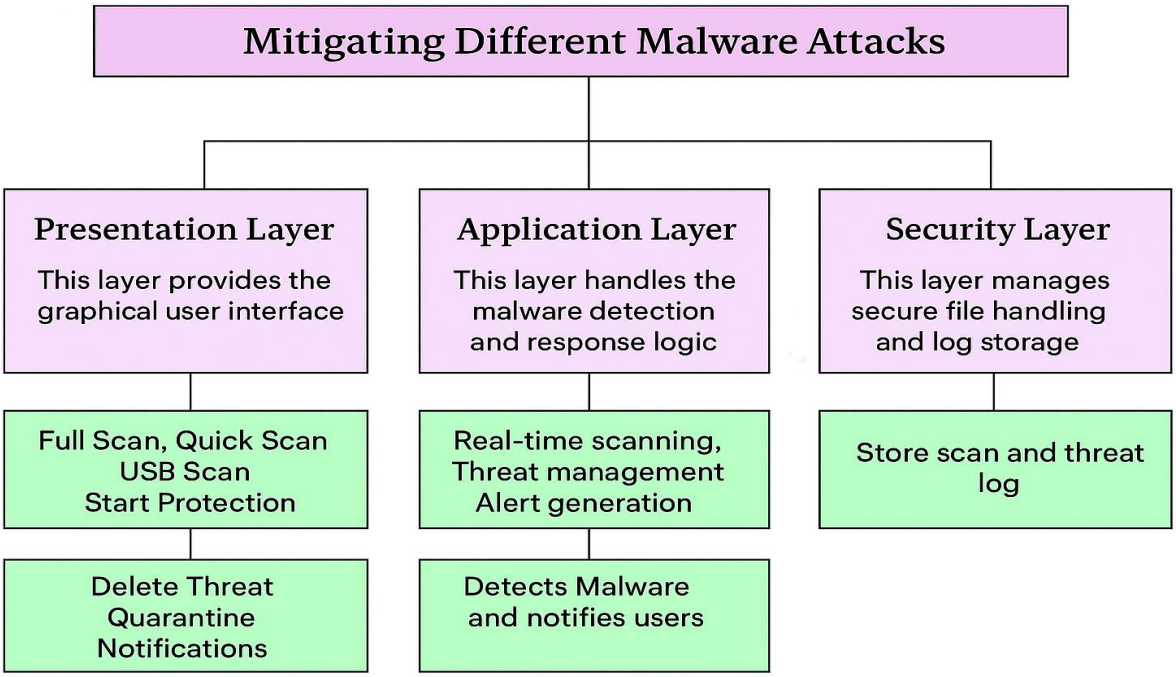

Figure 2. System Architecture Diagram

System Components and Modules

1.Malware Scanning Module

This module is responsible for scanning and detecting malware across the system and external drives. It allows users to perform different types of scans based on their needs.

Key Features:

- Full Scan – Scans the entire C: drive to detect all possible malware.
- Quick Scan – Focuses on critical areas such as documents, temporary files, and system folders.
- Folder Scan – Allows users to select specific files or folders to scan.

- USB Scan – Automatically detects and scans any connected flash drives for potential threats.
- Real-time Protection – Continuously monitors the system for malicious activity.

2.Threat Management Module

This module handles all detected threats and provides options for managing infected files.

Key Features:

- Delete Threats – Removes infected or malicious files from the system.
- Quarantine Files – Isolates suspicious files to prevent further spread while allowing later review.
- Restore Option – Safely restores files that are verified as false positives.
- Automatic Notifications – Alerts users of detected malware or high-risk threats in real time.

3.User Management Module

This module manages different user roles, permissions, and activities within the malware mitigation system.

Key Features:

- Admin: Can perform all scans, manage user accounts, configure security settings, and review detailed reports.
- Staff/User: Can perform quick, custom, or USB scans and respond to alerts or notifications.
- Activity Tracking: Logs all user actions for system transparency and accountability.

4.Reporting and Analytics Module

This module provides detailed insights, reports, and system statistics to help understand malware trends and system health.

Key Features:

- Detection Reports: Generates summaries of all detected threats, quarantined files, and deleted malware.
- Trend Analysis: Tracks recurring types of malware or infection sources (e.g., USB drives).
- System Performance Insights: Evaluates scan efficiency, false positives, and protection effectiveness.
- Email Reporting: Sends regular security updates and notifications to Zyro’s Pisonet administrators.

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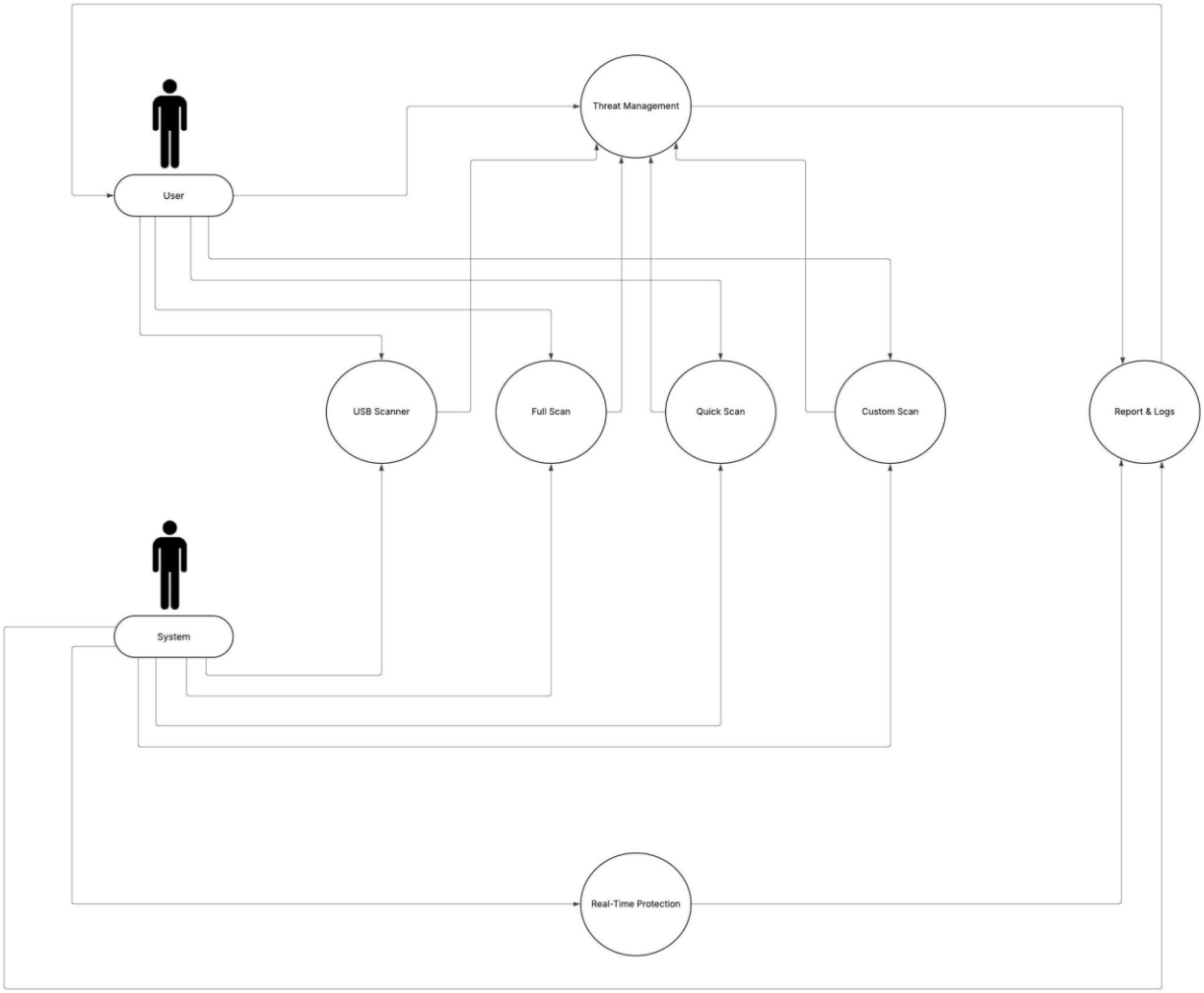
Use Case Diagram

Figure 3. Use Case Diagram



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Data Flow Diagram

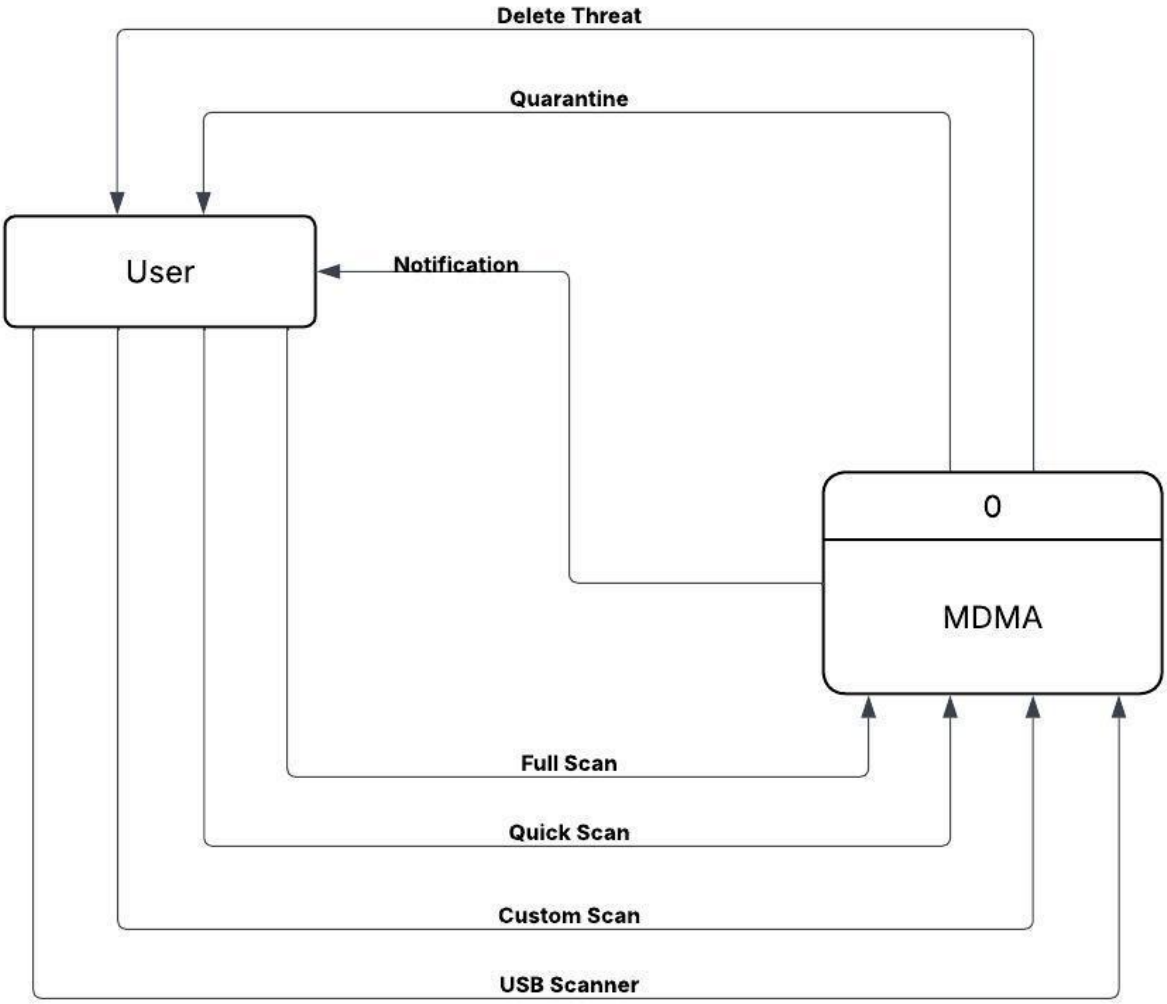

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Figure 4: Data Flow Diagram Level 0

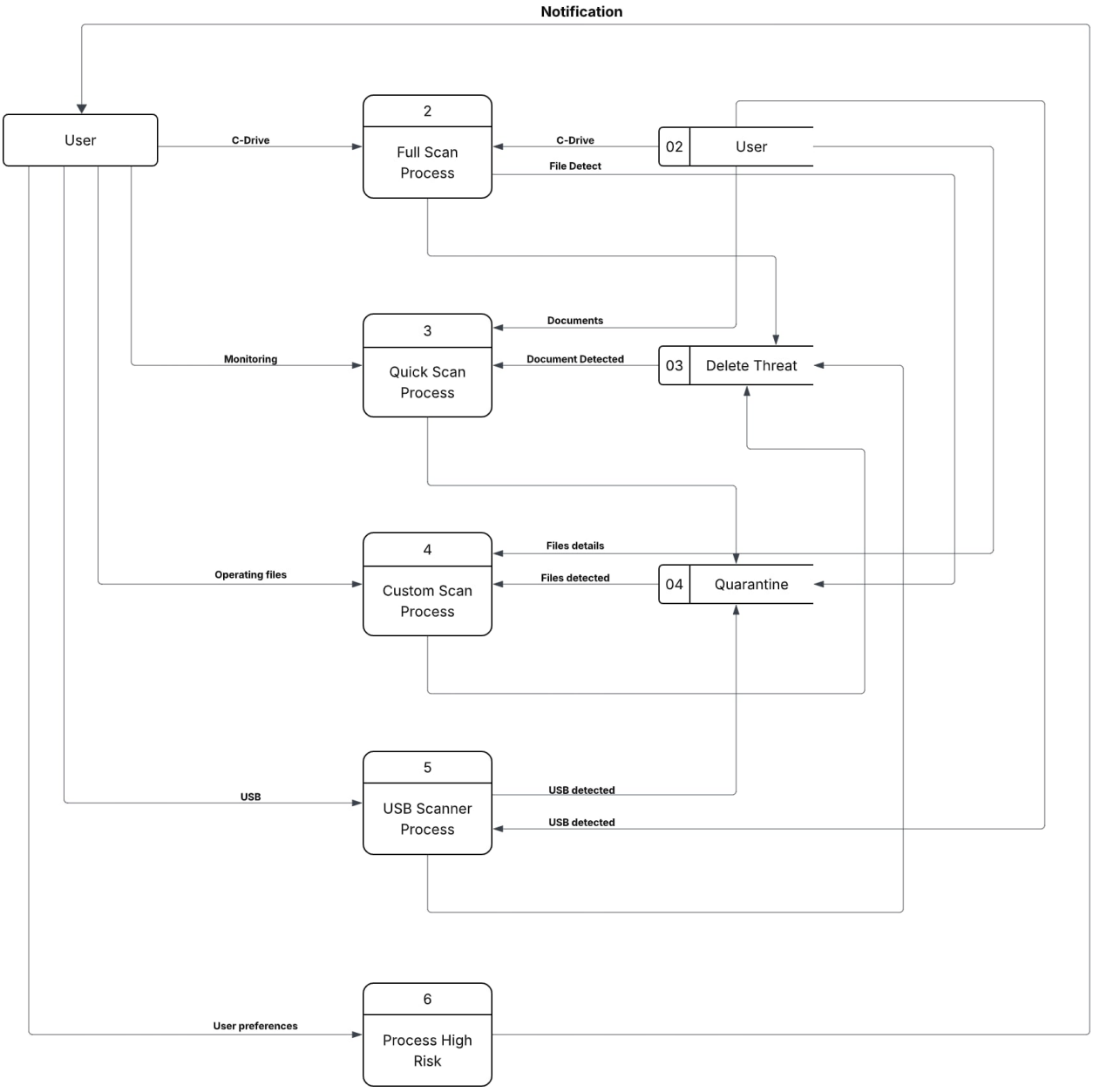

Figure 5: Data Flow Diagram Level 1

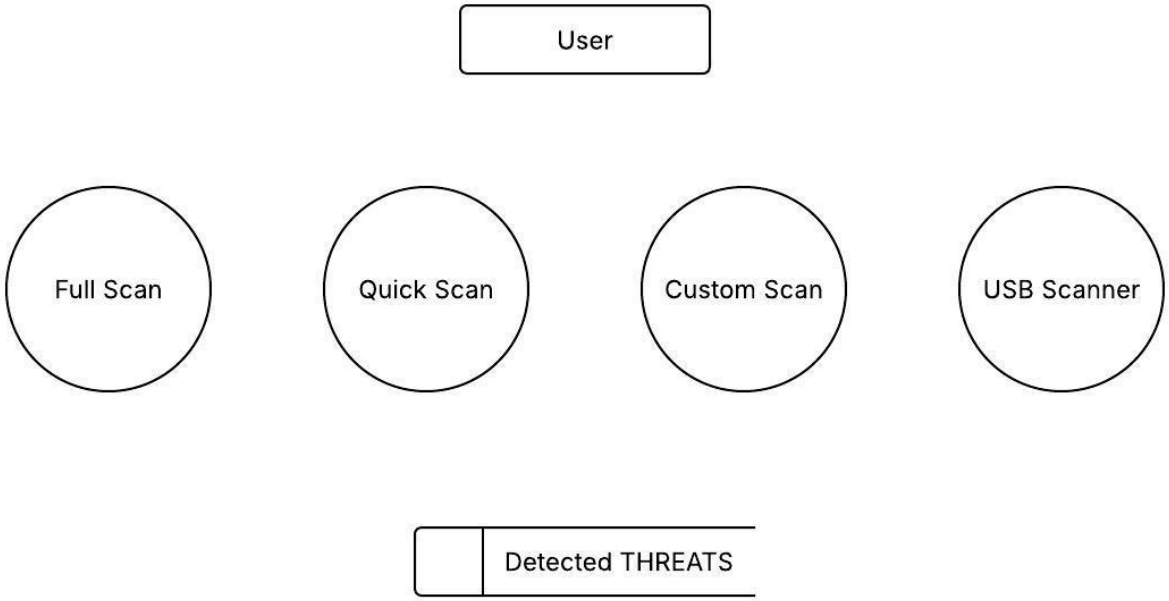

Figure 6: Data Flow Diagram Level 2

User Interface Design

The malware detection software was built with a modular, user-focused approach that emphasizes usability, accessibility, and clarity. Its main goal is to make malware detection, scanning, and threat management efficient and easy for both staff and clients at Zyro's Pisonet. Python's GUI frameworks, like Tkinter, were used to create a responsive and component-based interface. This allows for simple updates or reuse of elements such as the dashboard, scan controls, and threat management panels. The design provides a consistent, visually appealing layout that adjusts to different screen sizes, enabling smooth use on laptops and PCs.

A carefully chosen color scheme runs throughout the interface to give visual cues for threat severity and system status. Red indicates high-risk or detected malware threats, yellow points to active scanning processes or selected options, and green shows safe and operational status. This color coding helps users quickly understand system health and potential risks at a glance. The consistent use of icons, buttons, and hover effects creates a modern and engaging look while improving readability and user interaction.

When users log in, they see a display running in the background that offers a well-organized overview of the system. Key details, such as the number of scans completed, active threats, quarantined files, and system protection status, appear in visually distinct cards. Each card is color-coded and includes an icon to represent its category, allowing users to quickly assess security status. The main navigation panel, located on the left, includes Full Scan, Quick Scan, Custom Scan, USB Scan, Active Protection, and Settings, with the currently active option highlighted for easy recognition. Important functions like Delete Threat, Quarantine, and Notifications are placed prominently for quick access and immediate action.

The system provides real-time updates, showing scanning progress, detected threats, and system alerts dynamically on the dashboard. Users can move between different sections without delays, ensuring a smooth experience. Buttons, cards, and scan indicators include hover animations and visual feedback to enhance engagement and provide clear operational cues. Uniformity in font sizes, icon styles, and color use ensures a professional and cohesive

interface, giving users confidence and ease when managing malware threats.

Overall, the malware detection software application offers a strong, visually responsive, and user-friendly interface that makes system protection and threat management simple and effective. By combining clear visual indicators, intuitive navigation, and dynamic updates, the interface helps Zyro's Pisonet staff and clients confidently detect, manage, and neutralize malware threats.

Security Considerations

The following are the key security considerations implemented in the malware detection software to protect user systems, data, and ensure safe operations:

1. **Safe Handling of Files and Scans**
   All scanned files, including USB or system files, are handled securely to prevent accidental execution of malware. Suspicious files are quarantined in isolated directories, and malicious code is prevented from affecting the main system. Input validation is performed on file paths and user-provided data to mitigate risks from malicious scripts, ransomware, or trojans.

2. **Protection Against Common Attacks**
   The application includes safeguards against attacks such as malware injection, code execution exploits, and privilege escalation. By validating scan inputs, sanitizing file operations, and restricting actions based on roles, the system ensures that detected malware cannot compromise the application itself.

3. **Notifications and Alerts**
   High-risk threats trigger immediate notifications to users, ensuring timely action and preventing further spread. This proactive approach helps contain malware before it can affect other devices or networked systems.

Technology Stack

The Mitigating Different Malware Attacks application was developed using modern tools and technologies to ensure efficiency, security, and user-friendliness across all its modules.

- **Frontend** – The application's user interface was built using **Tkinter**, Python's standard GUI toolkit, to create a simple, clean, and responsive environment. Tkinter allows users to easily access various scanning options such as Full Scan, Quick Scan, Custom Scan, and USB Scan, as well as features for Active Protection, Quarantine, and Threat Deletion. A color-coded design system is implemented throughout the interface—red for high-risk threats, yellow for warnings, and green for safe system status—helping users quickly understand the security condition of their devices.

- **Backend** – The backend of the system is developed in Python, which handles all the core functionalities of malware detection, scanning, and real-time monitoring. It integrates two main detection engines: ClamAV for offline local scanning and VirusTotal API for online malware verification. This combination ensures accurate and comprehensive detection of various malware types, including viruses, trojans, worms, and spyware. The modular backend design allows for easy maintenance, system updates, and feature enhancements.

- **Additional Tools and Platforms** –
  - **ClamAV** – Serves as the main antivirus engine responsible for local malware scanning and detection.
  - **VirusTotal API** – Provides cloud-based scanning that compares files against multiple antivirus databases to ensure accurate threat identification.
  - **Visual Studio Code (VS Code)** – The primary integrated development environment (IDE) used for coding, debugging, and testing the Python scripts.
  - **PyInstaller** – Used to convert the Python project into a standalone executable (.exe) file for easy installation and deployment on Windows computers.

Data Gathering techniques

Survey Questionnaire for System Assessment

This survey is designed to assess the performance, usability, and overall satisfaction of our malware detection and protection system. The system includes key features such as malware scanning, detection of suspicious files, alert notifications, malware removal or quarantine, and a user-friendly interface for both mobile and desktop devices.

Your feedback is important and will help us enhance the system to better address various types of malware threats (e.g., viruses, Trojans, spyware, ransomware) and improve the user experience. Please answer each question truthfully and precisely based on your experience using the system.

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Demographic Information

Name: _____

Age: _____

1. The system effectively detects malware in the computer.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

2.The system quickly scans and identifies threats.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

3.The Full Scan feature works efficiently.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

4. The Quick Scan option is convenient for daily use.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

5. The USB Scan feature accurately detects threats from external drives.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

6. The Custom Scan feature is flexible and easy to use.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

7.The system provides reliable real-time protection.

☐ Very Satisfied

- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

8. Notifications for threats and alerts are clear and timely.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

9. The Quarantine feature works properly for isolating infected files.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

10. The Delete Threat function effectively removes detected malware.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral

- ☐ Dissatisfied
- ☐ Very Dissatisfied

11. The user interface is simple and easy to navigate.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

12. The system does not slow down the computer while scanning.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

13. The system enhances overall computer security.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

14. The application performs consistently without crashes or errors.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

15.The installation and setup process were easy to follow.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

16.The system meets the security needs of Zyro’s Pisonet Computer Shop.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

17.The system helps reduce the risk of malware infections in the shop’s computers.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

18.The overall performance of the system is reliable.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

19.The system provides good value for its purpose.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

20.Overall, I am satisfied with the *Mitigating Different Malware Attacks* system.

- ☐ Very Satisfied

- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

Sampling Methods for Respondents

A simple random sampling method will be used in this study. Ten individuals who completed the survey and are regular users of Zyro's Pisonet computers will be the respondents. This method was chosen to ensure the participants are actual users who frequently interact with the computer systems. They can provide relevant and accurate feedback about the malware detection application.

Even though the study is limited to ten respondents, this sampling method allows for a focused and representative group. It minimizes bias and ensures that each user has an equal chance to share their experiences and views. Through this approach, the study aims to gather reliable and meaningful data that show how users perceive the Mitigating Different Malware Attacks software application in terms of its usability, functionality, and effectiveness in protecting computers and USB devices from potential malware threats.

Data Analysis Techniques

The survey data were examined using both numbers and descriptive methods to assess how effective, usable, and necessary the Mitigating Different Malware Attacks software application is. The main goal was to see how well the system could detect, prevent, and manage different types of malware threats on Zyro's Pisonet computers. The respondents included regular users and staff who shared their thoughts through structured surveys based on their experiences with the system.

For the quantitative analysis, spreadsheet tools like Google Sheets and Microsoft Excel were used. Survey responses were entered, organized, and processed through coding procedures. Basic statistical tools like mean, percentage, and frequency distribution helped identify the most common answers and levels of agreement among users. Visual tools like bar graphs and pie charts clearly displayed the findings, making it easier to understand user perceptions of the application's reliability, functionality, and ease of use.

Qualitative data from open-ended survey questions were analyzed to spot recurring themes and insights about the application's performance, user interface design, and ability to detect threats. This mixed-method approach combined statistical data with direct user feedback for a complete evaluation. The results showed that developing the Mitigating Different Malware Attacks application is very relevant and beneficial, providing a solid solution for protecting computers and USB devices from various malware threats while keeping system performance and user convenience intact.

Ethical Considerations

The Mitigating Different Malware Attacks software application was developed with a strong focus on ethical standards and data privacy regulations to protect user information and maintain the system's integrity. The project follows the Data Privacy Act of 2012 (Republic Act No. 10173), which protects sensitive and personal data from misuse, alteration, and unauthorized access. All collected information, including user feedback and system testing results, was kept confidential and used only for academic and system development purposes.

During the survey and evaluation processes, we respected participants' privacy and confidentiality. We did not record or disclose any personally identifiable data. All responses were anonymized to ensure fair analysis. Participants were informed about the purpose of the study and provided their voluntary consent before taking part in any testing or feedback activities.

To further uphold our ethical responsibility, we processed survey data using secure spreadsheet tools that did not reveal individual identities. These steps guaranteed that all information remained confidential and was used solely to evaluate the effectiveness, accuracy, and usability of the malware detection system.

Overall, the project emphasizes transparency, user privacy, and ethical integrity throughout its development and evaluation. This ensures that the Mitigating Different Malware Attacks system operates responsibly while safeguarding users from possible data and security risks.

Chapter 4

Results and Interpretation

This chapter presents a thorough look at the results from developing and implementing the Mitigating Different Malware Attacks software. It shows how the system effectively tackled common issues in malware prevention, such as slow detection, lack of real-time protection, and limited user awareness, by offering a modern, automated, and interactive malware detection solution. The results came from a series of system tests, performance evaluations, and user feedback sessions with regular users and staff at Zyro's Pisonet. The findings showed that the system significantly improved malware detection efficiency, increased computer protection, and lowered the risk of infection from internal storage and external USB devices. By integrating ClamAV and VirusTotal API, the system accurately detected various types of malwares. The Tkinter-based GUI provided an easy-to-use interface for users with different technical skills. Through structured testing, users reported greater confidence in the system's ability to scan, quarantine, and remove threats effectively. Features like real-time protection, USB scanning, and quarantine management ensured that malware threats were identified and dealt with before they could harm the system. The application's alerts and notifications also played a vital role in keeping users informed about detected risks and system activities. This chapter discusses how the system's performance matched the project's original goals, especially in improving security awareness, speeding up threat response time, and maintaining device stability. A comparison with traditional antivirus methods highlighted the system's advantages in terms of speed, flexibility, and integration with both local and cloud-based scanning engines.

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Task Manager

Gantt Chart / Project Timetable – Mitigating Different Malware Attacks

WEEK	TASK	UI/UX	BACKEND	QA	DOCUMENTER
Week 1–2	Project Planning & Requirement Gathering	_____	_____	_____	Gather project requirements, identify malware detection objectives, and create system proposal.
Week 3–4	User Interface Design & Wireframing	Design layout using Tkinter (buttons, panels, scan logs, and progress bars).	_____	_____	Document wireframes, GUI structure, and design concepts.
Week 5–8	Frontend Development	Develop the main interface for full scan, quick scan, custom scan, and USB scan options.	_____	_____	Record progress and update documentation of interface design.

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Week 9–12	Backend Development	_____	Develop core malware detection features using ClamAV, integrate VirusTotal API for online scanning, and implement system monitoring tools like Watchdog.API routes	_____	Document backend setup, integration procedures, and API utilization.
Week 13–16	System Integration	Link Tkinter GUI with backend scanning logic and detection results.	Finalize real-time protection, quarantine, and log storage features.	_____	Document integration and workflow between interface and detection modules.
Week 17–20	Testing & Debugging	Refine GUI responsiveness and usability.	Debug scanning logic, VirusTotal communication, and file handling processes.	Conduct malware simulation tests and verify detection accuracy.	Record test cases, issues found, and applied solutions.
Week 21–22	System Evaluation	Improve interface design based on user	Optimize ClamAV scan speed and VirusTotal API	Retest system stability and performance under	Summarize evaluation results and testing

		feedback.	requests.	different conditions.	reports.
Week 23-24	Documentation Finalization	————	————	————	Compile and finalize project documentation, screenshots, and test summaries.
Week 25-26	Presentation & Submission	Prepare demonstration of the interface and scanning functions.	Showcase backend performance and malware detection results.	Assist during final system testing and demo session.	Finalize project deliverables and submit documentation.

Presentation of Results

The Mitigating Different Malware Attacks software application was developed, implemented, and evaluated to improve the security of computer systems. It detects, analyzes, and mitigates various types of malwares. The system focuses on providing real-time protection, scanning flexibility, and accurate detection of potential threats across local drives and external devices like USB flash drives. With its easy-to-use interface and multi-engine detection approach, the system aims to reduce infection risks and raise user awareness of system security.

During testing and evaluation, the system underwent thorough assessments of its performance and functionality. These assessments measured its ability to identify malware, scan multiple files efficiently, and maintain stable system operations. The ClamAV engine was tested for its accuracy in local detection, while integration with the VirusTotal API offered extra verification through cloud-based multi-engine scanning. The system's responsiveness and scanning speed were evaluated under various conditions, including full system scans, quick scans for specific folders, and real-time protection scenarios.

User testing involved regular users and staff from Zyro's Pisonet. They assessed the system's usability, interface clarity, and overall functionality. Users performed tasks like detecting malware in USB devices, isolating threats in the quarantine section, and monitoring live protection features. Feedback indicated that the application was easy to navigate, provided clear notifications about detected threats, and gave accurate results without slowing down the system.

Data collected from testing logs and user input were analyzed to evaluate the system's accuracy, efficiency, and user satisfaction. Results indicated that the system successfully detected and isolated malware files, prevented further spread with active protection, and maintained stable performance even during continuous operation. The clear and organized Tkinter-based interface boosted usability, allowing users to monitor scans, view logs, and manage quarantined files effectively.

Impact of Mitigating different malware attacks on the Information Technology Field

The Mitigating Different Malware Attacks software has greatly improved how computer systems protect against harmful software and cyber threats. Traditionally, malware detection relied on manual virus checks or single-engine antivirus tools, which often missed new or complex threats. By using automation, real-time monitoring, and multi-engine scanning, this project represents a key improvement in proactive cybersecurity solutions. By combining ClamAV for local detection and the VirusTotal API for cloud-based verification, the system creates a hybrid approach that boosts malware detection accuracy and system reliability. This integration allows for a broader scanning process, ensuring that even zero-day or less common threats are identified and dealt with effectively. Additionally, the use of Tkinter for the graphical user interface (GUI) makes the software application more user-friendly. Both technical and non-technical users can easily run scans, check results, and manage quarantined files. The project also impacts the IT field by promoting cybersecurity awareness and preventive protection. Its real-time scanning and USB protection features target common infection sources that are often ignored in public computer settings, like those in computer shops and shared workstations. By finding and isolating potential threats before they can spread, the system reduces downtime, data corruption, and system damage, which are major concerns in IT operations and digital maintenance.

Contribution to Technological Advancements

The Mitigating Different Malware Attacks software application plays a key role in improving technology by adding modern cybersecurity tools and automated protection to everyday computer use. It features an intelligent, user-friendly system that ensures

security, accuracy, and efficiency in detecting and managing malware threats. By automating scanning processes and providing real-time protection, the project decreases manual effort and strengthens computer defense. This minimizes downtime and keeps system performance steady. To improve protection and flexibility, the system uses advanced technologies like ClamAV for local malware detection, the VirusTotal API for multi-engine verification, and Tkinter for its interactive graphical interface. These tools work together to create a secure, user-friendly, and efficient environment. Users can perform full scans, quick scans, USB scans, and custom file checks with ease. The system combines cloud-based and local scanning methods, showing how open-source and real-time technologies can create more reliable and accurate malware detection systems. Additionally, the project encourages the use of automated and proactive security measures to prevent the spread of harmful files and strengthen digital resilience. With active protection, quarantine features, and continuous monitoring, the system supports safe computing practices and highlights the importance of user awareness in cybersecurity.

Project Structure

```
Project Root |
├── app.py
├── convert.txt
├── guide.txt
├── requirements.txt |
├── Quarantine
| └── CLAMAV |
├── clamav.lib
├── clamav_rust.lib
```

```
├── clamd.exe
├── clamscan.exe
├── freshclam.exe
├── sigtool.exe
├── README.md
├── COPYING.txt
├── freshclam.conf
├── freshclam.log |
├── conf_examples
| └── clamd.conf.sample |
```

Chapter 5

Summary, Conclusion, and Recommendation

Summary

The goal of this system was to improve computer security and protect users from different types of malwares. This was achieved by designing and developing the Mitigating Different Malware Attacks software application for Zyro’s Pisonet Computer Shop. This system can detect, analyze, and stop malicious software from infecting computers through both internal storage and external devices like USB drives. By offering multiple scanning options, including Full Scan, Quick Scan, Custom Scan, and USB Scan, the

application increases protection, flexibility, and user control in spotting potential threats within the computers at Zyro's Pisonet.

To ensure timely detection, the project used effective malware detection methods and a user-friendly interface built with Tkinter. This interface allows users to easily access important features like scanning, quarantine management, and active protection. The system was developed through a structured process that included gathering requirements, design, coding, testing, and evaluation to ensure it is reliable and performs well.

The results showed that the software application effectively identifies and reduces malware threats, lowering the risk of infection and system compromise. Users and administrators at Zyro's Pisonet reported that the tool was easy to use, dependable, and efficient at scanning files and drives for potential threats. The system met its main goals of providing real-time protection, accurate detection, and an easy experience for users, making it a reliable solution for preventing malware attacks and keeping system integrity within the computer shop.**Conclusion**

The results showed that the *Mitigating Different Malware Attacks* software application developed for Zyro's Pisonet Computer Shop successfully met its goal of strengthening computer security and preventing malware infections. The system effectively detected and mitigated harmful software, ensuring that all computers and external devices remained protected from potential threats. By providing multiple scanning options and real-time protection, the

application helped maintain the safety and reliability of the company's computer units used by customers.

Overall, the project achieved its primary objective of protecting Zyro's Pisonet systems from various forms of malware, reducing the risk of data corruption, and ensuring continuous, secure operation of their computers. This initiative reflects the company's commitment to delivering a safe and dependable computing experience for all its users.

Recommendations

Based on the development, implementation, and evaluation of the *Mitigating Different Malware Attacks* software application, the following recommendations are proposed to enhance the system's functionality, efficiency, and security for **Zyro's Pisonet Computer Shop**:

1. **System Expansion**
   Future versions of the system should extend its functionality to cover a wider range of devices and platforms. At present, the system primarily focuses on local drives and USB scanning. Expanding its capability to include network monitoring and protection across multiple computers would strengthen security within the entire computer shop. This improvement would allow the system to detect and contain malware that could spread through shared connections or local networks.

2. **Feature Enhancement**
   Although the current version already includes key features such as Full Scan, Quick Scan, Custom Scan, and USB Scan, it is recommended to add additional tools such as scheduled

automatic scanning, detailed scanning reports, and improved quarantine management. Providing customizable notifications and real-time threat alerts would also enhance user awareness and allow for quicker responses to detected malware.

3. **Strengthen System Security**
   To maintain the reliability and safety of the system, stronger security measures should be implemented. These include encrypting stored scan logs, securing system configurations, and adding multi-factor authentication for administrative access. Regular updates of malware definitions and system patches are also essential to ensure protection against newly emerging threats. Conducting regular maintenance checks and vulnerability assessments will further guarantee that the system remains secure and efficient.

4. **Comprehensive Testing and Performance Evaluation**
   It is recommended that future testing involve deployment across multiple computers to evaluate the system's speed, accuracy, and performance under real-world operating conditions. Stress testing should also be conducted to ensure that the system remains stable and responsive during large-scale scanning or when multiple users are using the application simultaneously. This process will help identify potential issues and improve the system's scalability and reliability.

References (APA Format)

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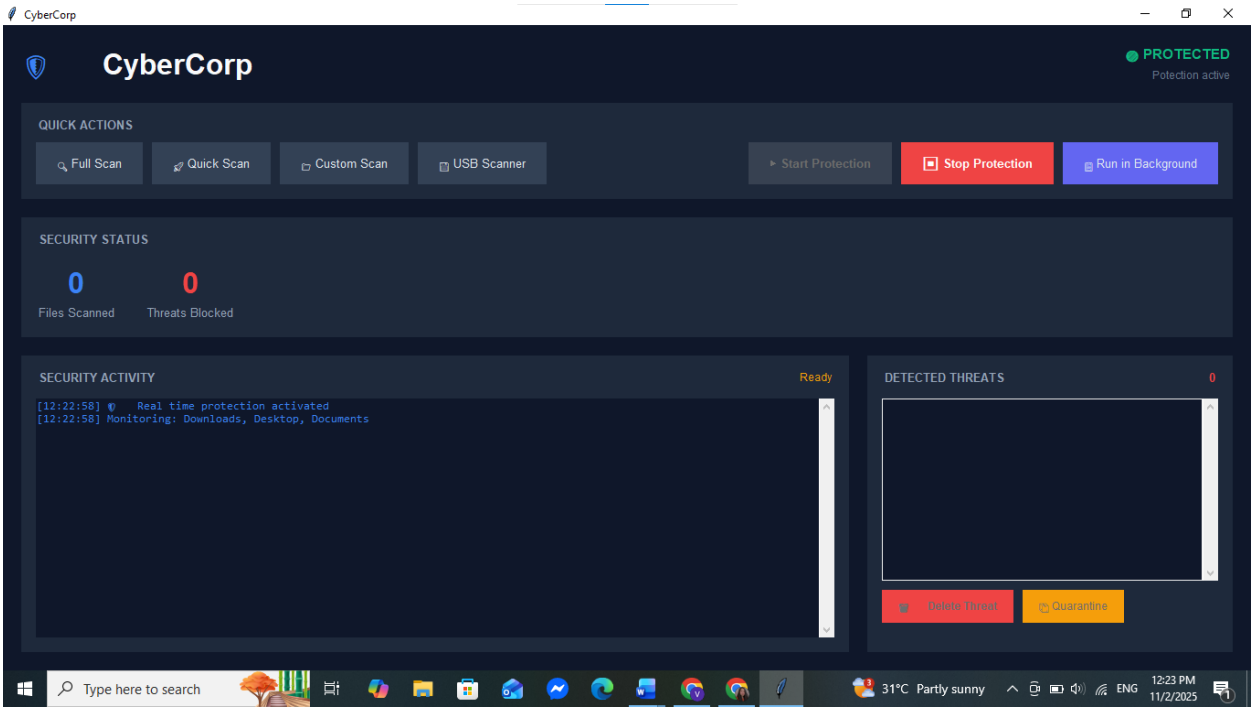
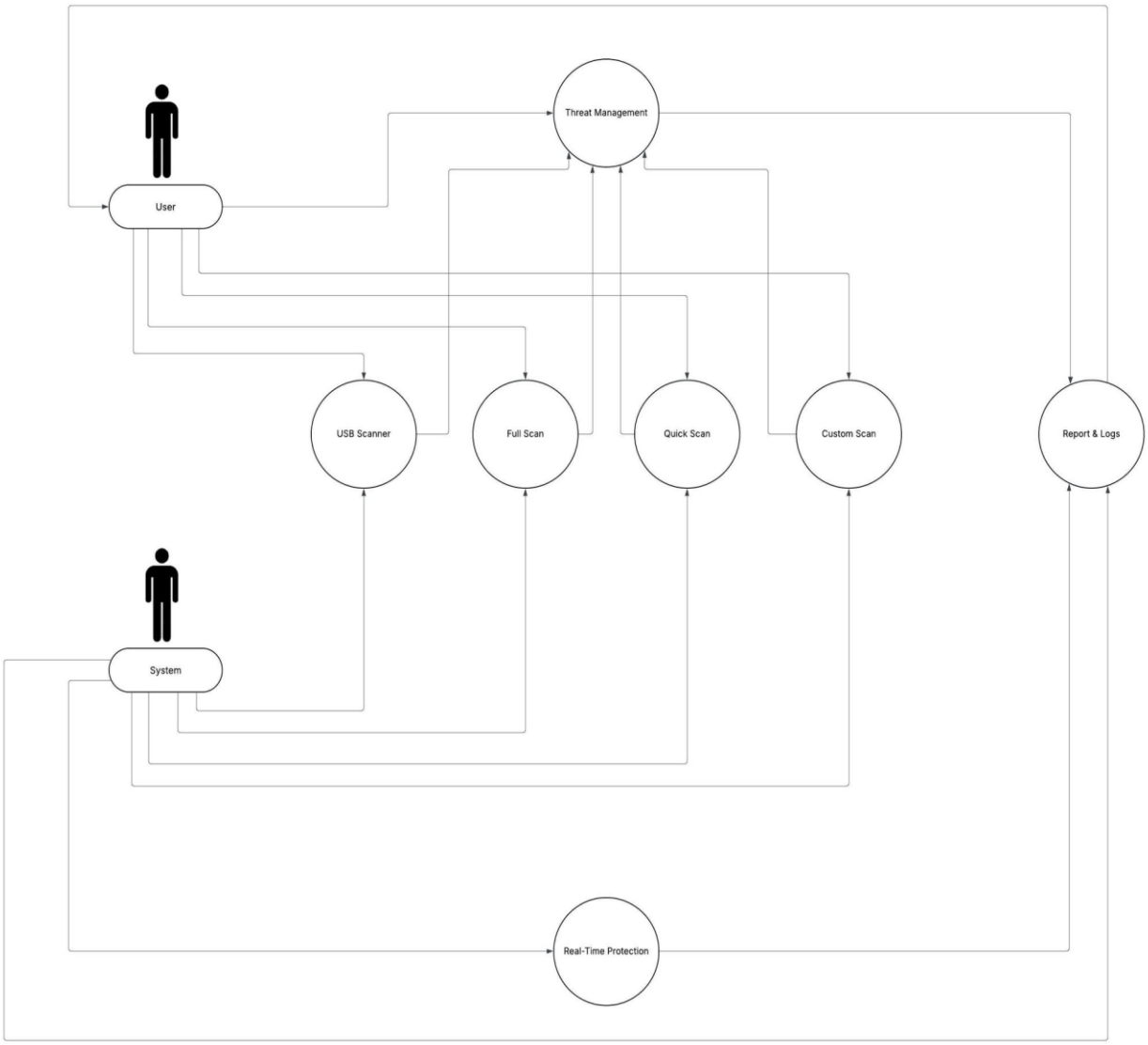
Appendices:

Appendix A: System Interface Screenshots



Appendix B: Use Case Diagram



Appendix C: Data Flow Diagram

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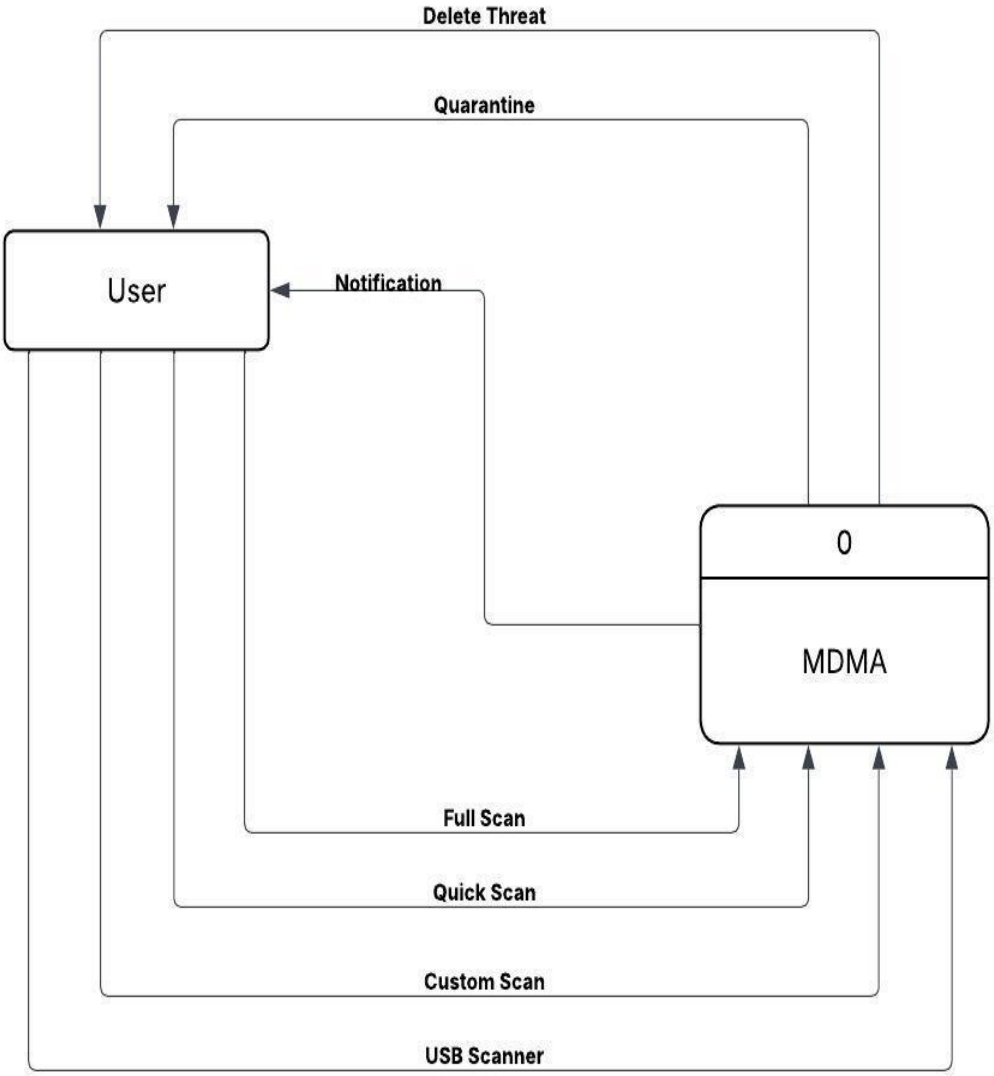

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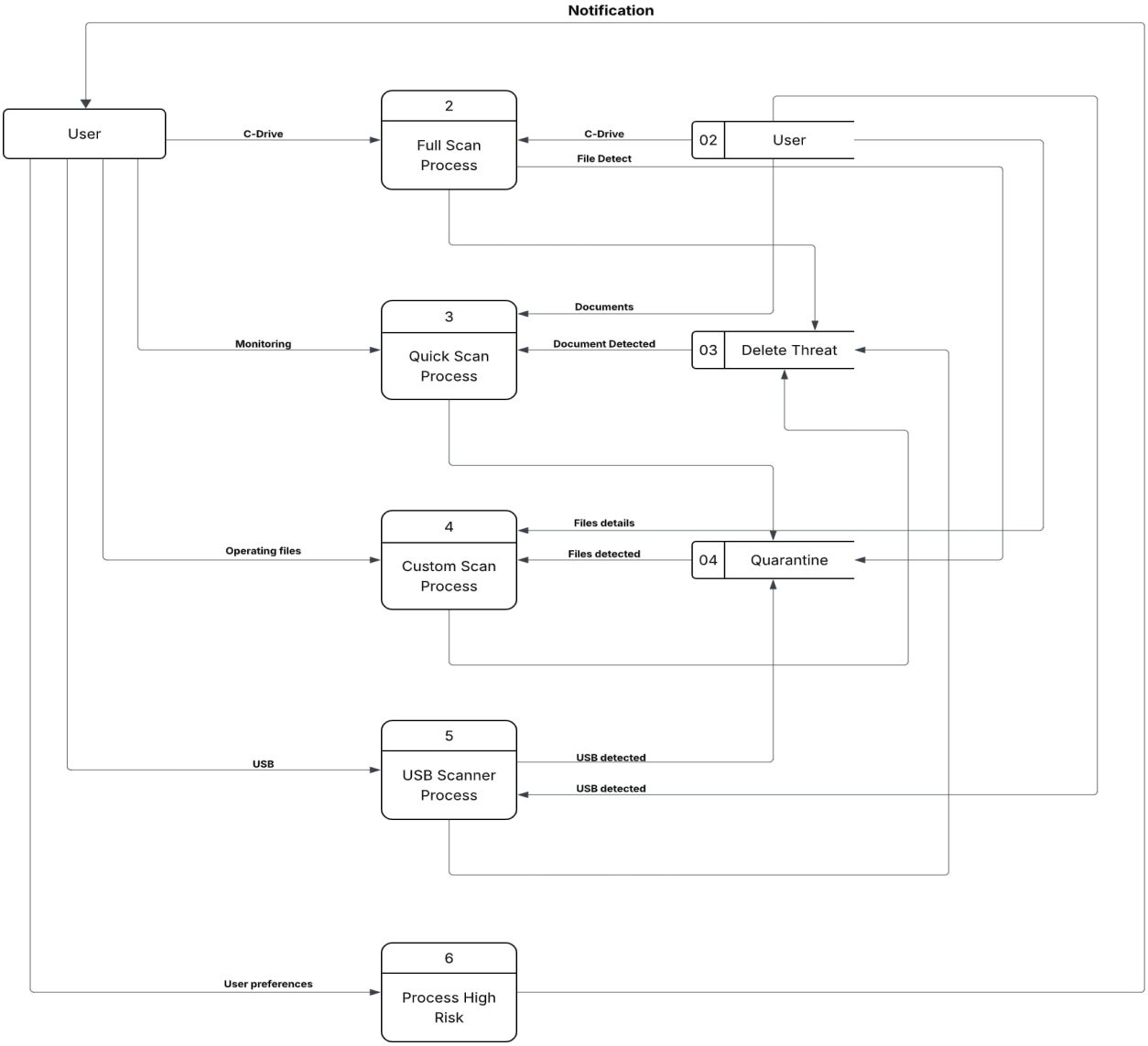

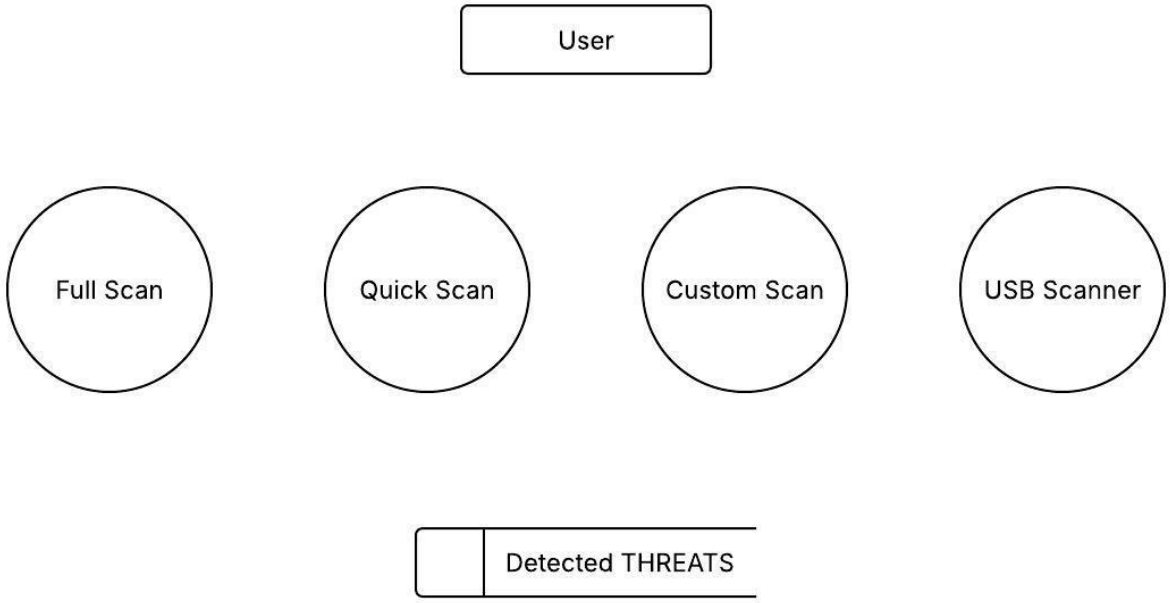

Appendix D: User Survey Form and Summary

Demographic Information

Name: _____

Age: _____

1. The system effectively detects malware in the computer.

   ☐ Very Satisfied

   ☐ Satisfied

   ☐ Neutral

   ☐ Dissatisfied

   ☐ Very Dissatisfied

2.The system quickly scans and identifies threats.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

3.The Full Scan feature works efficiently.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

4. The Quick Scan option is convenient for daily use.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

5. The USB Scan feature accurately detects threats from external drives.

- ☐ Very Satisfied
- ☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

6. The Custom Scan feature is flexible and easy to use.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

7.The system provides reliable real-time protection.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

8.Notifications for threats and alerts are clear and timely.

☐ Very Satisfied

☐ Satisfied

☐ Neutral

☐ Dissatisfied

☐ Very Dissatisfied

9.The Quarantine feature works properly for isolating infected files.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

10. The Delete Threat function effectively removes detected malware.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

11.The user interface is simple and easy to navigate.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

12.The system does not slow down the computer while scanning.

- ☐ Very Satisfied
- ☐ Satisfied

- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

13. The system enhances overall computer security.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

14. The application performs consistently without crashes or errors.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

15. The installation and setup process were easy to follow.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied

- ☐ Very Dissatisfied

16.The system meets the security needs of Zyro's Pisonet Computer Shop.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

17.The system helps reduce the risk of malware infections in the shop's computers.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

18.The overall performance of the system is reliable.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied

- ☐ Very Dissatisfied

19. The system provides good value for its purpose.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

20. Overall, I am satisfied with the *Mitigating Different Malware Attacks* system.

- ☐ Very Satisfied
- ☐ Satisfied
- ☐ Neutral
- ☐ Dissatisfied
- ☐ Very Dissatisfied

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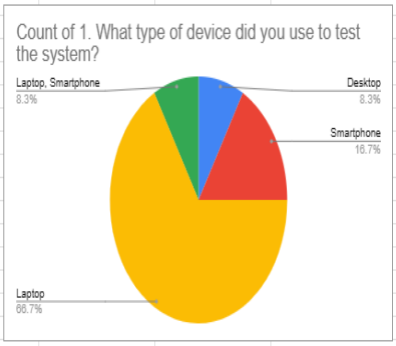
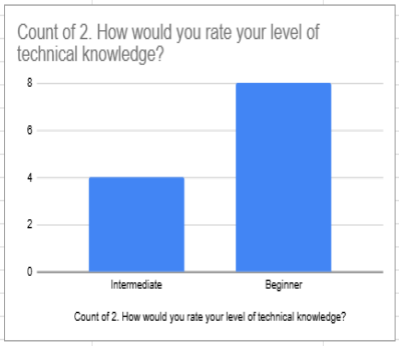
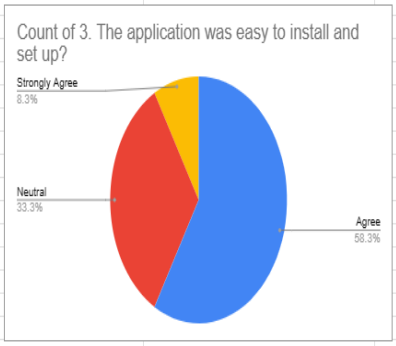
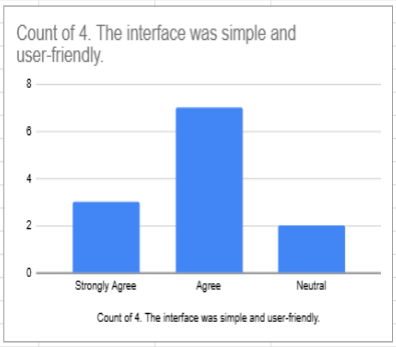
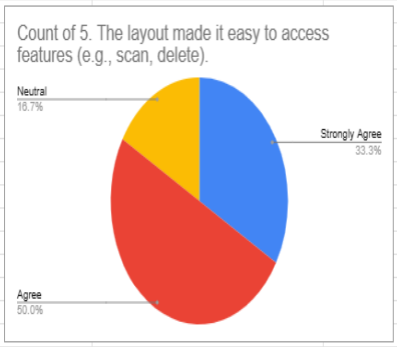
- Summary of Responses (e.g., tables, charts)

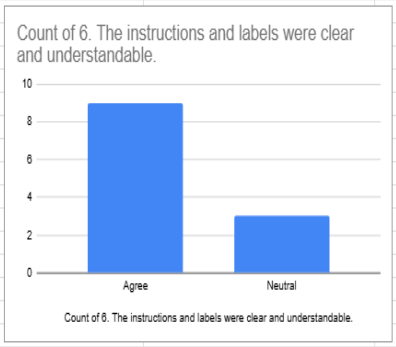












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